

1. Basic Conceptual Questions

Q1. What does the **Error column** actually represent?

Answer: Error = Actual Demand – Forecast.

+ve error = **Under-forecasting** (you ordered too little → risk of stockout).

–ve error = **Over-forecasting** (you ordered too much → excess inventory & holding cost).

Q2. What is the difference between **Error** and **Absolute Percentage Error (APE)**?

Answer: Error keeps the sign (+ or –) → used for Bias.

APE removes the sign and converts to % → used for MAPE.

Q3. Why do we calculate **Bias** separately from MAPE?

Answer: MAPE tells “how big is the average mistake?”.

Bias says “does the model have a **bad habit** of always being too high or too low?” (systematic error).

2. Calculation Questions (based on YOUR file)

Q4. In your Excel, why is 3MA MAPE calculated only from April–December (not January)?

Answer: In the first 3 months (Jan–Mar), 3MA has no previous 3 months → forecast = 0 (invalid).

We only use periods where the model actually produced a meaningful forecast.

Q5. Calculate MAPE manually for **June** for both models (show steps).

3MA: Actual=200, Forecast=131.67 → Error=+68.33 → APE = $|68.33|/200 = 34.17\%$

ES: Actual=200, Forecast=138.75 → Error=+61.25 → APE = 30.63% (approx.)

Q6. Your data shows ES MAPE = **15.02%** and 3MA = **22.25%**. Is this improvement “meaningful”? Why?

Answer: Yes — 33% reduction in error + Std. Dev. dropped from 40.72 to 29.78 (27% less variation).

In the supply chain, even 5% better MAPE can save lakhs in inventory cost.

3. Interpretation & Insight Questions (Most Important)

Q7. Both models have **positive bias** (+3.33 and +4.32). What does this mean for inventory planning?

Answer: The models are **consistently under-forecasting**.

→ You will keep facing stockouts, especially during peaks (June).

Action: Either increase safety stock or tune the models higher.

Q8. The Day 2 insight says “Not all accuracy improvements are meaningful—look for consistency.”

Explain with your two models.

Answer: ES has lower MAPE **and** much lower Std. Dev. of errors (29.78 vs 40.72).

This means ES errors are more predictable → easier to plan purchase orders, production, and safety stock.

A model with slightly lower MAPE but wild swings is actually worse in real life.

Q9. If Actual demand in one month is zero, what problem occurs with MAPE?

Answer: Division by zero → MAPE becomes undefined.

That’s why we sometimes use **MAE** or **WMAPE** in real projects.

4. Real-Life Supply Chain Application Questions

Q10. You are the Supply Chain Manager. Your boss asks: “Which model should we use for next year’s planning in SAP?”

What will you reply and why? (Use your Best model selection table)

Answer: “We should use **Exponential Smoothing ($\alpha=0.5$)**.

It has 15.02% MAPE vs 22.25% for 3MA, lower variation, and reacts faster to trend changes.

This will reduce both stockouts and excess inventory.”

Q11. If demand suddenly becomes very seasonal (like festival spikes), which metric becomes more important — MAPE or Bias?

Answer: Bias becomes critical because a model that under-forecasts during peak season will cause huge lost sales.

MAPE alone can hide this.

Q12. Your MAPE is 15%. Is this good or bad?

Answer: Depends on the product!

- Stable FMCG (toothpaste) → 15% is acceptable.
- New launch / fashion / electronics → 15% is actually **very good**. Always compare with industry benchmark + business impact (cost of error).

Quick Revision Tip

Print or screenshot the table below and keep it handy:

Concept	What it tells you	Good Value in your file	Problem if bad
MAPE	Average % error	ES = 15.02%	High → costly mistakes

Bias	Systematic over/under	+3 to +4 (mild)	High → always stockout
Std. Dev. of Errors	Consistency / predictability	ES = 29.78 (better)	High → planning nightmare
